Week 1 - Introduction to R

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15 October 2026

What is R?

- Programming language focused on statistics
- Accessible with no programming experience
- Can perform every step of research process
- Industry standard in quantitative research
- Free and highly customizable

What is RStudio?

- Integrated Development Environment (IDE)
- Think MS Word for programming with R
- Useful features:
  - Good editor with autocomplete, spell checker, etc.
  - Shortcuts for common commands + automatic formatting
  - Execute code directly in RStudio
  - Useful data manager and viewer

What are packages?

- Additional functions for specific uses
- Easy management directly through R
- Install once with `install.packages("packagename")`
- Load what you need with `library(packagename)`

What is RMarkdown?

- R creates code - useful but not presentable
- Word etc. has no good functions to show code
- RMarkdown to combine code, documentation, and outputs
- Based on Markdown - a "markup language" (text formatting)
- "Knit" to create presentable HTML or PDF files

Projects and Scripts

What is an R Project?

- Linked to a specific folder
- Stores plots, tables, data etc. in project folder
- Allows working with different scripts

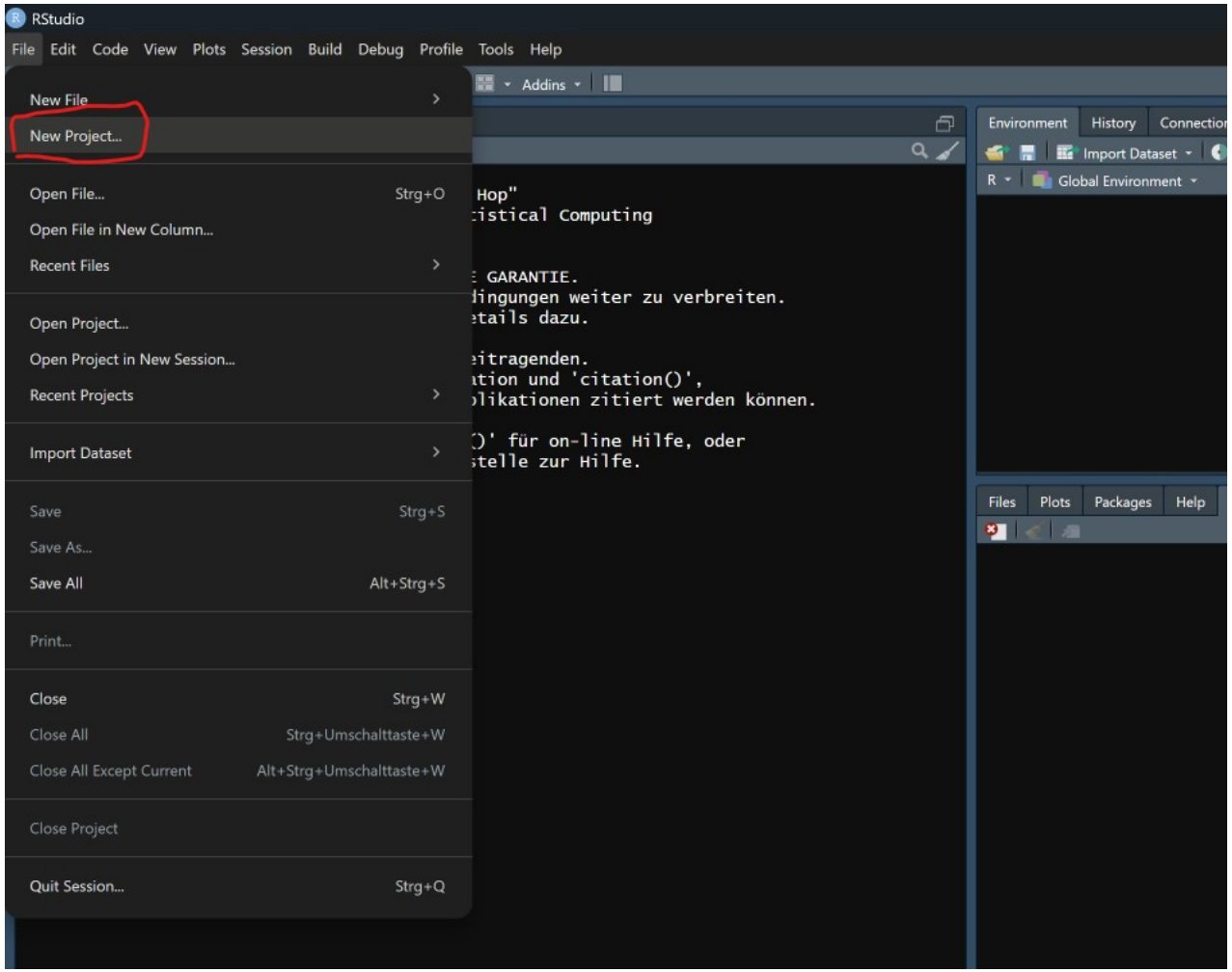
What is an R Script?

- File with R code that you can run
- Typically have different scripts for different tasks

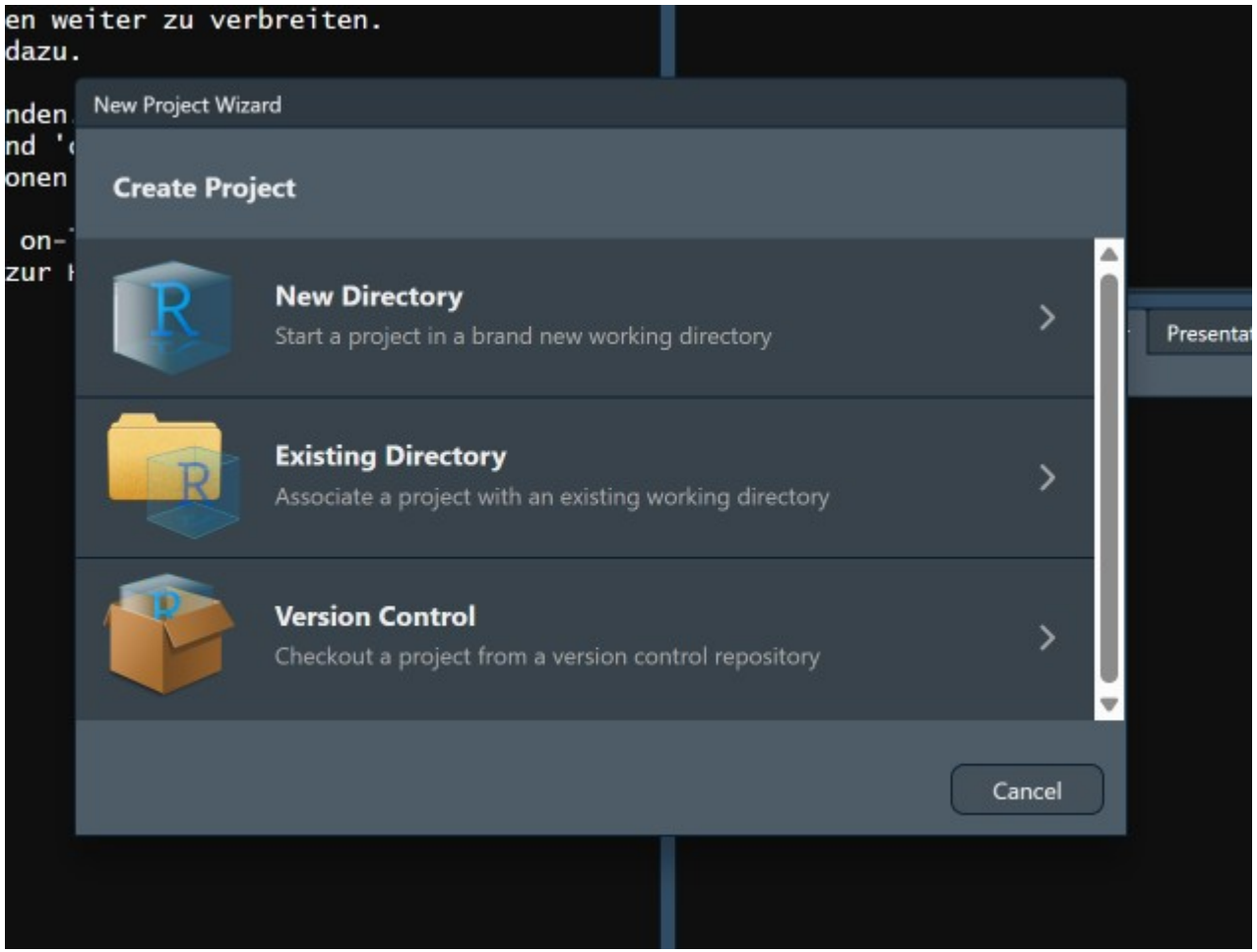
A Typical Workflow

- Open a new project
- Open a new script
- Load necessary packages
- Load data and start your tasks

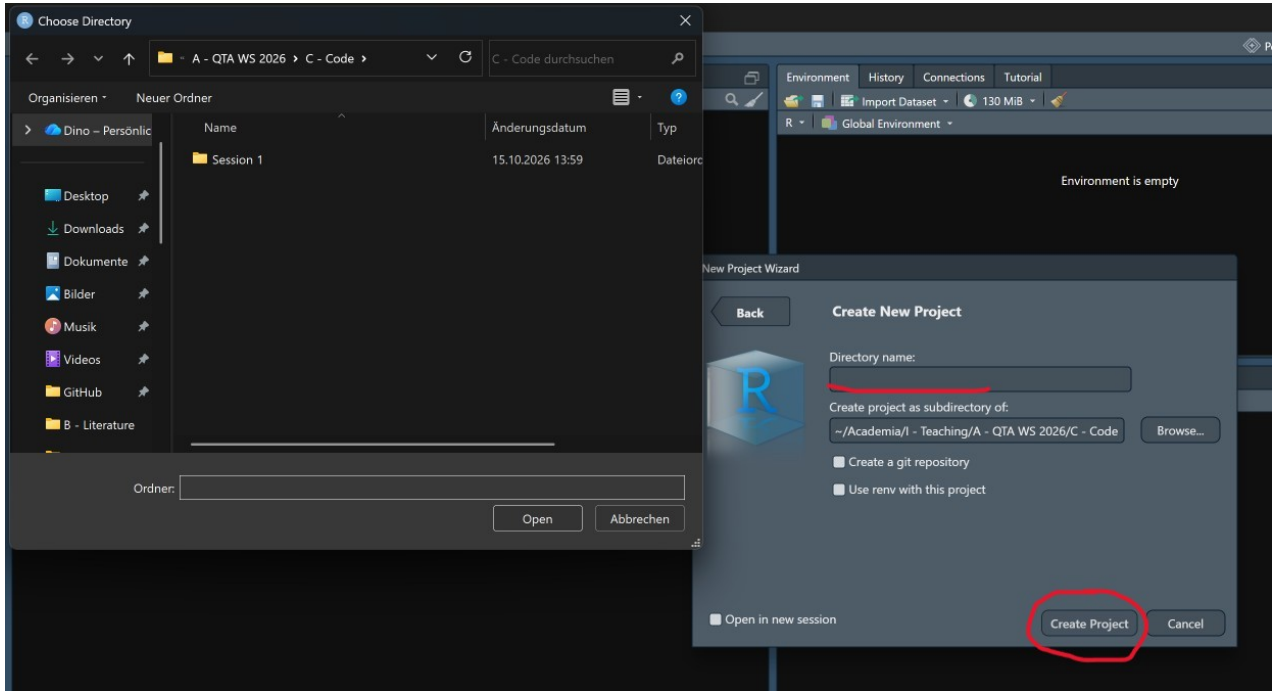
Creating a project



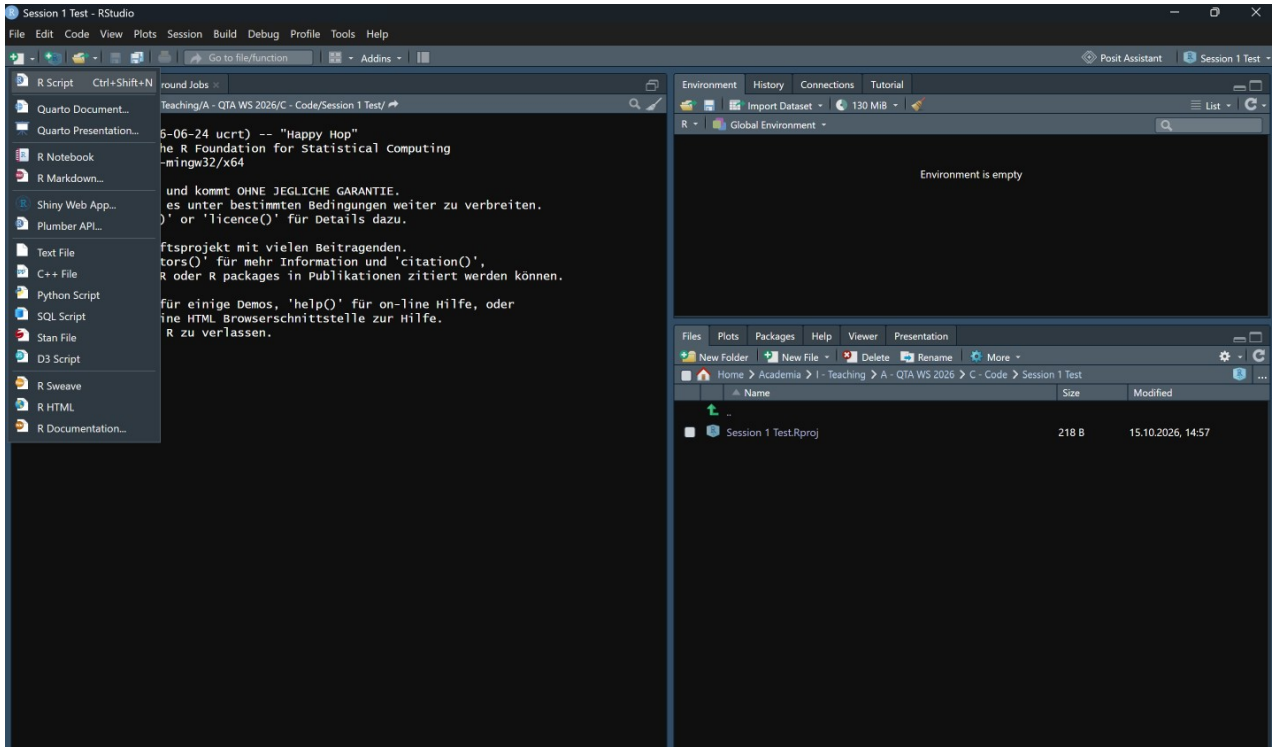
Creating a project



Creating a project



Creating a script



Panes of RStudio

- Top left: Script $\Rightarrow$ this is where you code
- Bottom left: Console $\Rightarrow$ run code "on the fly"
- Top right: Data manager $\Rightarrow$ see all your variables
- Bottom right: Multi-use panel
  - Files: Shows the files in the project folder
  - Plots: Shows plots you generate (see future sessions)
  - Packages: Shows what you have installed
  - Help: Access function or package help